

	A	B	C
17			
18	Data		Formula
19	n	64	
20	sample mean	20	
21	sample standard deviation	3	
22	confidence coefficient(1- α)	0.95	
23	alpha(α)	0.05	=1-B22
24	$\alpha/2$	0.025	=B23/2
25	Z $\alpha/2$	1.959963985	=ABS(NORMSINV(B24))
26	standard error of mean	0.375	=B21/SQRT(B19)
27			
28	calculation of 95% C.I for mean:		
29	lower confidence limit=	19.26501351	=B20-B25*B26
30	upper confidence limit=	20.73498649	=B20+B25*B26
31			
32	Hence required 95% confidence limit for mean is (19.26501351,20.73498649)		
33			

	A	B	C
17			
18	Data		Formula
19	n	1000	
20			
21	P	0.23	
22	Q	0.77	=1-B21
23	confidence coefficient(1- α)	0.99	
24	α	0.01	=1-B23
25	$\alpha/2$	0.005	=B24/2
26	Z $\alpha/2$	2.575829304	=ABS(NORMSINV(B25))
27	standard error of P	0.013307892	=SQRT((B21*B22)/B19)
28			
29	99% confidence limit for proportion:		
30	Lower confidence limit=	0.195721141	=B21-B26*B27
31	Upper confidence limit=	0.264278859	=B21+B26*B27
32			
33	In percentage terms: 19.57% to 26.43%		
34			